

From DataSight to HydroServer

How the Nile Basin Initiative moved its Regional Hydrological Monitoring Data Management System (DMS) to Open Source

What we will cover



1

Context

The basin, NBI in Early Warnings for All, the network and DataSight

4 min

2

Why we moved

Cost, control, fragmentation, and the road to a decision

4 min

3

How we migrated

Model first, three tracks, and what went wrong

5 min

4

Results

HydroServer in production, visualization, quality control and the open API

4 min

5

Lessons and thanks

What we would tell another users

3 min

A shared basin, a shared system

The Nile Basin Initiative(NBI)

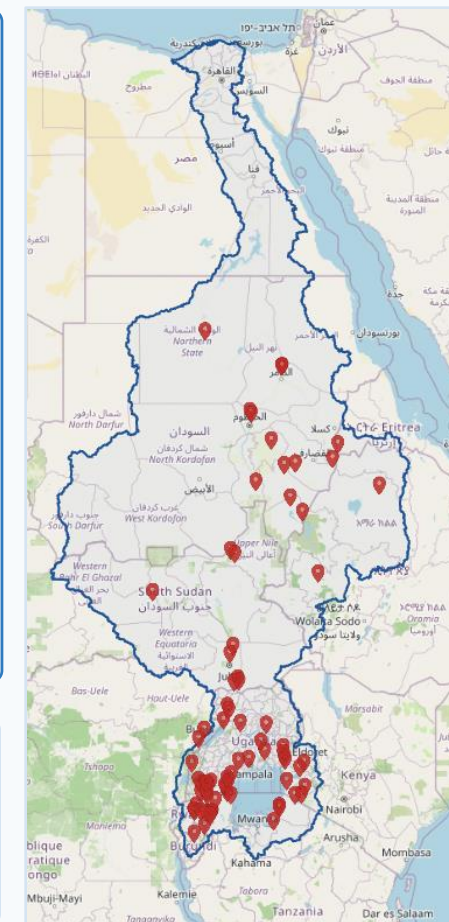
- The regional intergovernmental partnership of the Nile river riparian states
- Burundi · DR Congo · Egypt · Ethiopia · Kenya
- Rwanda · South Sudan · Sudan · Tanzania · Uganda
- Cooperating on sustainable management and development of shared Nile Basin water resources

What the monitoring system supports

- Basin-wide water resources planning
- Flood and drought forecasting
- Transboundary water allocation dialogue
- Evidence-based decisions by Member States
- ...

THREE KINDS OF MONITORING, ONE REGIONAL RECORD

Hydrometeorological, water quality and groundwater data come from different instruments, manufacturers and national arrangements. The regional platform turns them into one consistent database.



Where NBI fits in Early Warnings for All



The UN Early Warnings for All (EW4All) initiative rests on four pillars. NBI contributes most directly to Pillar 2, and to Pillar 3 through regional partners.

PILLAR 1

Disaster risk knowledge

Led by UNDRR

PILLAR 2

Detection, observation, monitoring, analysis and forecasting

Led by WMO

NBI · STRONGEST FIT

PILLAR 3

Warning dissemination and communication

Led by ITU

NBI · WITH ICPAC and NBD

PILLAR 4

Preparedness and response capabilities

Led by IFRC

Pillar 2 — the core of what NBI does

- Hydrological observations across the basin
- River-flow forecasting
- Flood and drought monitoring
- Basin-level water information shared across Member States

This work sits directly in the detection, monitoring and forecasting part of EW4All.

WITH REGIONAL PARTNERS

Pillar 3 — reaching communities

NBI also works with regional organizations such as Nile Basin Discourse (a Civil Society Organization – connecting NBI’s activities with NB communities), ICPAC (IGAD Climate Prediction and Applications Centre) on disseminating warnings to communities through mobile technologies.

Where NBI fits in Early Warnings for All

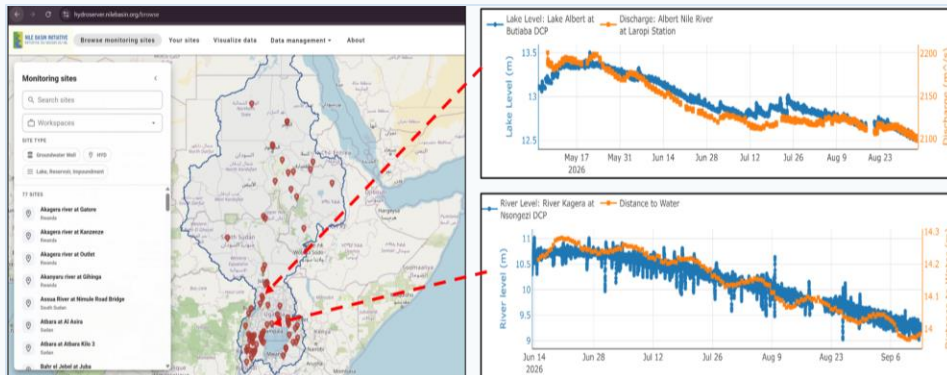
EW4All Pillar 2 — Detection, observation, monitoring, analysis and forecasting

Pillar 2 — the core of what NBI does

- Nile Basin Regional Hydrological Monitoring System with observations across the basin (River Water Levels and Flows, Lake Water Level, Rainfall Amount, River and Lake Water Quality and Groundwater Levels).
- Basin-level water information shared across Member States
- Nile Basin River Forecasting System -River-flow forecasting and hindcasting
- Nile River Flood Forecasting System
- Eastern Nile Flood Forecasting System
- Co-generation of Nile Basin Seasonal Hydrological Outlook and Advisory to Member States through the Nile Basin (REWG) on Hydrology forum.

This work sits directly in the detection, monitoring and forecasting part of EW4All.

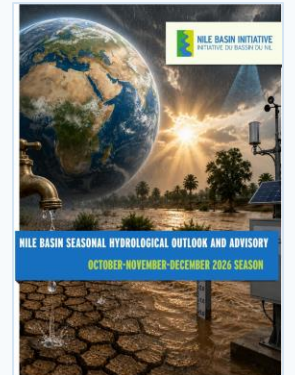
1 · MONITORING AND FORECAST



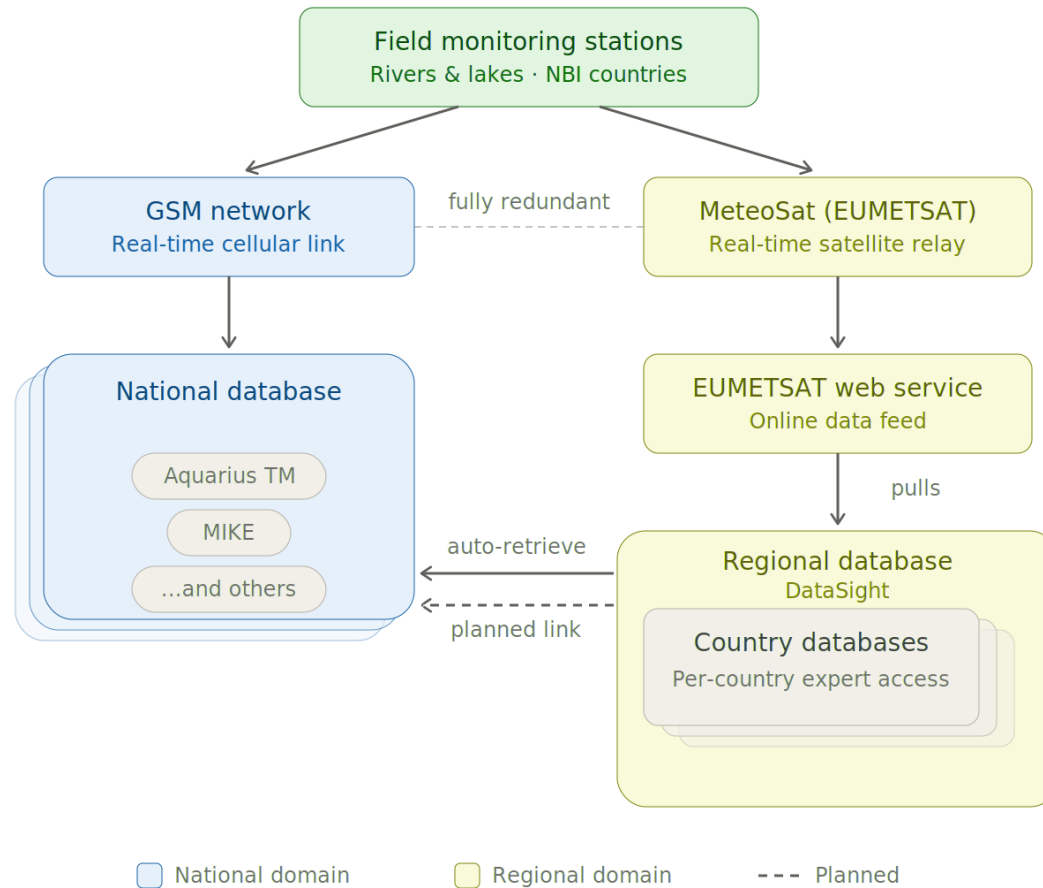
2 · EXPERT ANALYSIS & INTERPRETATION



3 · DISSEMINATION



Hydromet system architecture — DataSight (before migration)



How data reaches DataSight

- Field stations transmit over a redundant real-time link GSM cellular and MeteoSat (EUMETSAT) satellite relay
- The GSM pathway lands in a national database (Aquarius TM, MIKE and others)
- The regional database on DataSight retrieves data from the EUMETSAT web service

TWO DOMAINS

National domain: each country's own database and instruments.

Regional domain: DataSight, with per-country database for expert access., some national databases already retrieve from the regional database.

Where the pressure appeared

DataSight did its job for three years. It was replaced because cost, control and extensibility were not sustainable.

COST

Recurrent licensing

- Fees recur every year
- A burden for institutions with constrained budgets
- Cost grows with scope, not with value

CONTROL

Vendor dependence

- Closed source, single-vendor control
- Roadmap and support tied to the contract(SMA)
- Proprietary data model creates lock-in

CAPABILITY

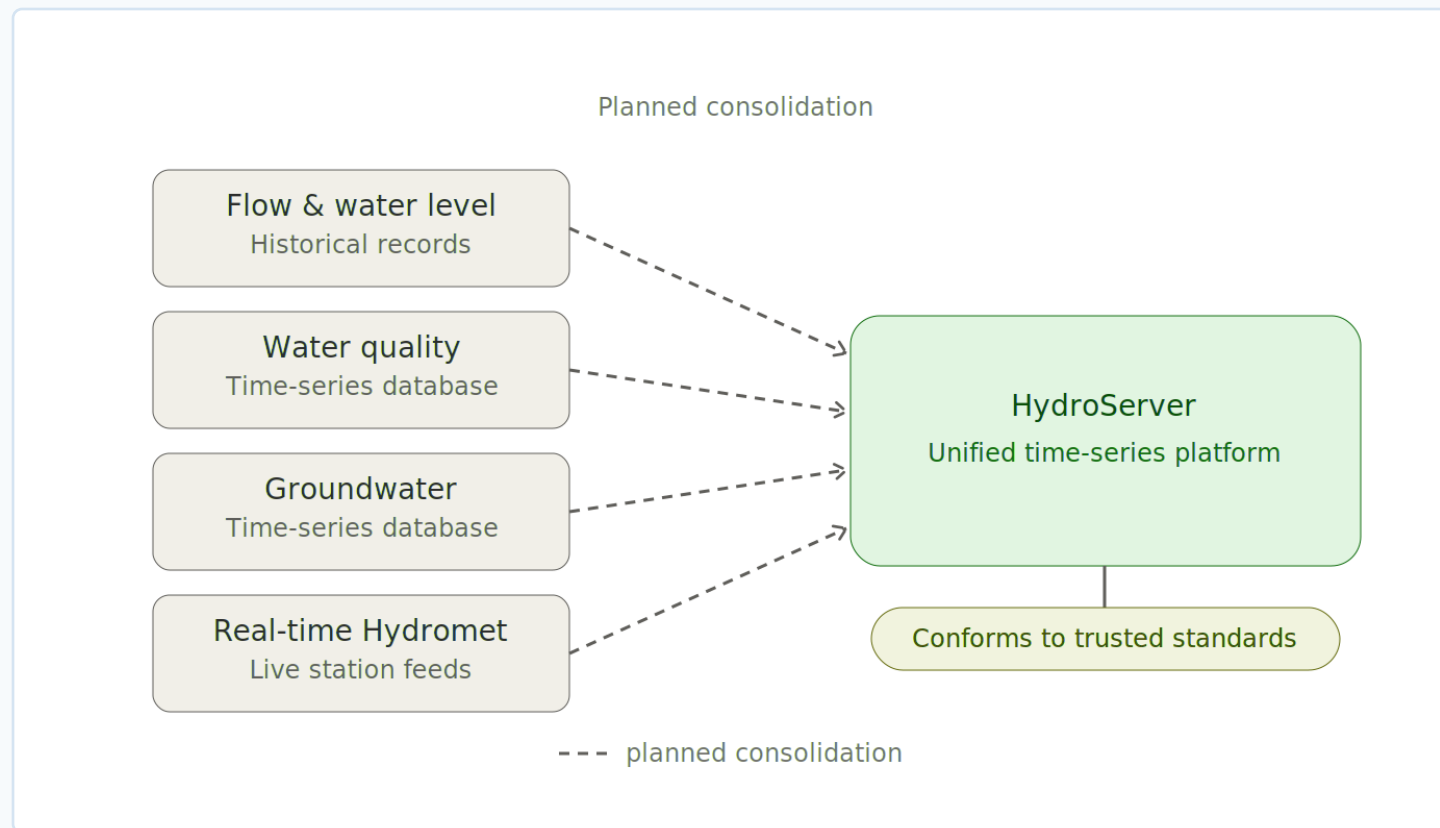
Extensions needed

- Dashboards and applications required development
- Limited interoperability with national systems
- Each extension a new negotiation

AND THE DATA WAS FRAGMENTED

Hydrometeorological, water quality and groundwater data each sat in its own time-series system , **a separate license and separate administration for each, at extra cost.** Bringing them onto one platform mattered as much as the license fees.

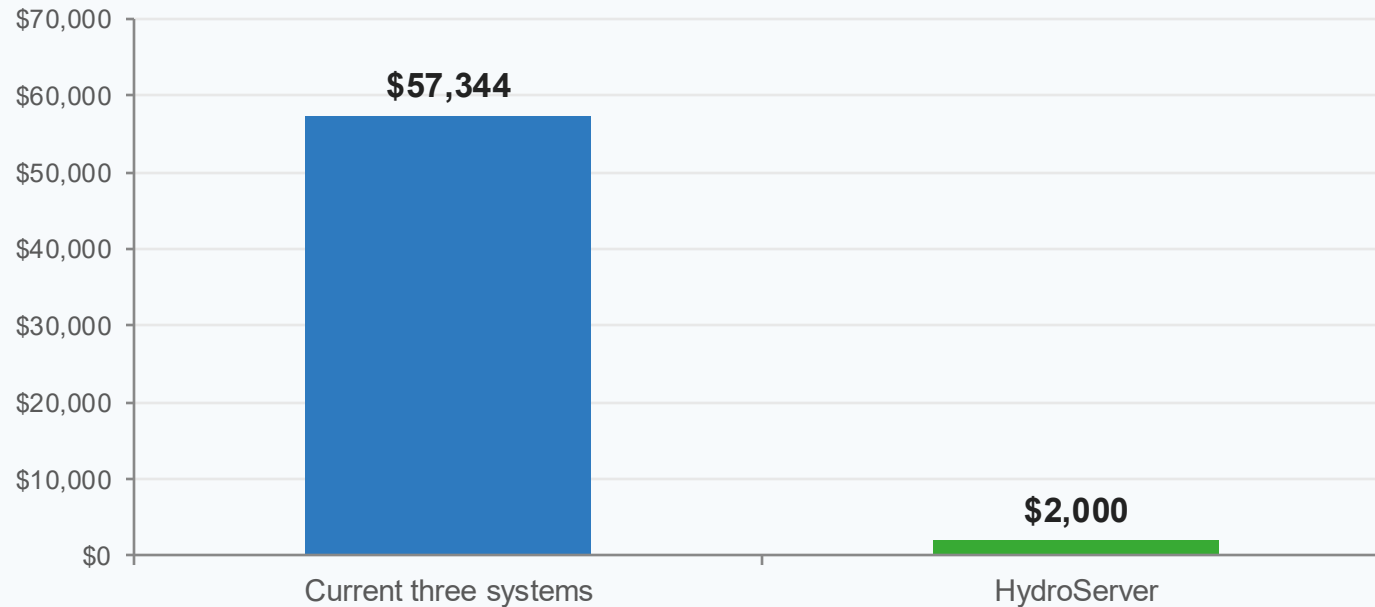
Time-series database consolidation



THE PLANNED CONSOLIDATION

Flow & water level, water quality, groundwater and real-time hydromet feeds — each its own time-series database — **consolidate into HydroServer as one unified platform, conforming to trusted (open) standards.**

What consolidation is worth



Annual licence / hosting cost, all data types

~96%
lower annual cost

SAVED EACH YEAR

≈ US\$55,000

redirected from licences to monitoring, capacity building and operations.

Figures from the NBI feasibility assessment presented to the Hydrological Working Group, June 2026.

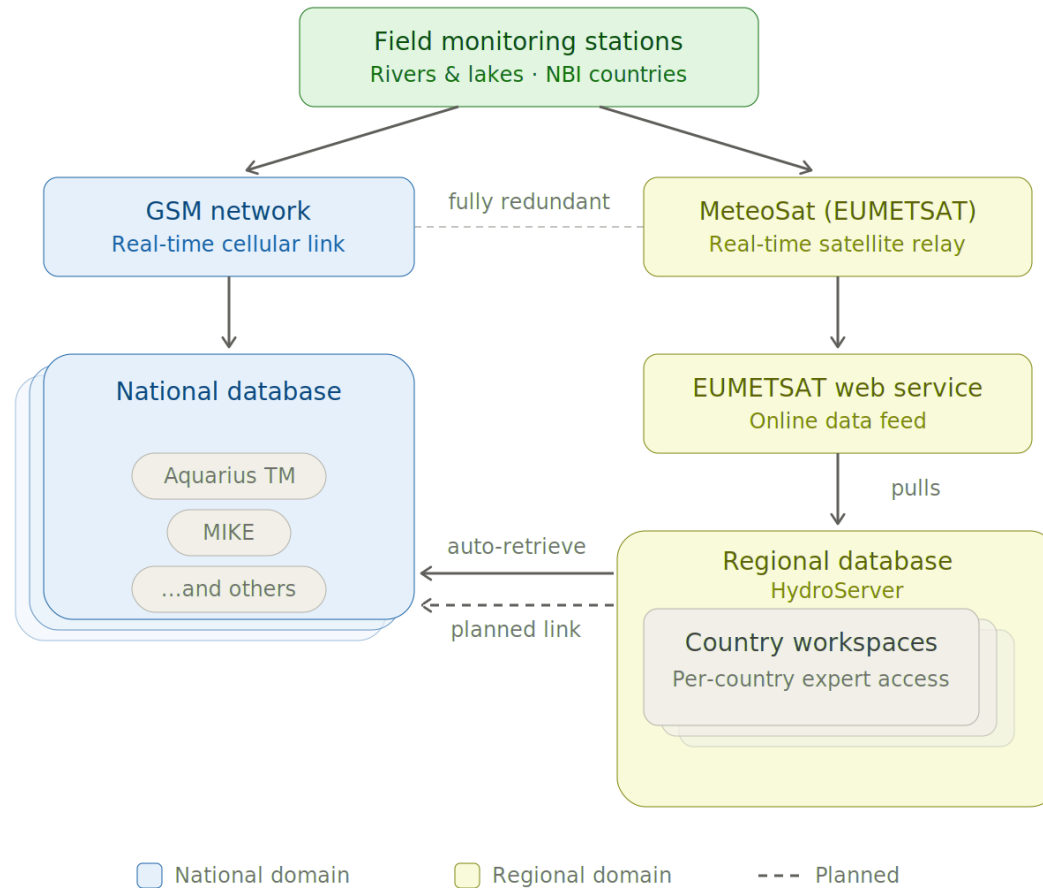
From assessment to a live regional system



LOW RISK BY DESIGN

The pilot was tested without disrupting any country's existing monitoring: **nothing changed in the field or at national level.**

Hydromet system architecture — HydroServer



What actually changes

- Field stations, telemetry (GSM / MeteoSat) and national databases stay exactly as they are
- Only the regional node changes: DataSight is replaced by HydroServer
- Per-country access becomes workspaces instead of DataSight's country databases

WHY THIS MATTERS FOR MIGRATION

Nothing in the field or at the national level needed to change. The transition was a swap at the regional database only, which allowed the pilot to be tested without disturbing any country's existing monitoring.

Migrate the model, not just the numbers

A DATASIGHT EXPORT

Timestamps and values

That is usually all an export contains.



WHAT MUST ALREADY EXIST IN HYDROSERVER

Before a single observation can be uploaded

Site

Method

Observed Property

Unit

Processing Level

Get the mapping right once per station

Every future upload, manual or automated, just works.

METADATA FIRST

This is why migration starts with metadata, not with data.

Three migration tracks

1

Metadata setup

Define Sites, Sensors and Datastreams

2

Historical backfill

Upload the DataSight archive

3

Pipeline cutover

Switch live feeds to HydroServer

Tracks 1 and 2: UI and API

- Sites and sensors were created manually through the UI
- Datastreams and archive import were automated with Python scripts through the API, using the hydroserverpy library

Track 3: only when verified

- Cut over a station only once Tracks 1 and 2 are verified for it

Challenges, and how we solved them

Four challenges from NBI's own migration, and how each one was solved.

QC RANGES

Static range on a cumulative counter

A rising battery-charge counter was capped at 200; every value above it was dropped for weeks, with the sensor working the whole time.

THE FIX

For counters, set the QC maximum far above any plausible lifetime total.

TIMESTAMPS

No timezone on timestamps

Ambiguous local times caused silent multi-hour offsets in the stored data.

THE FIX

Always include a UTC offset on every timestamp before upload.

BATCH UPLOADS

Duplicate timestamps in one batch

A bulk upload rejected the entire batch (HTTP 409).

THE FIX

De-duplicate by timestamp, keeping the last value, before uploading.

CUSTOMIZATION

Simple changes needed code

Branding and small adjustments could only be made in code, so NBI had to maintain a private repository.

THE FIX

Simple customization belongs in settings, not in code. Raised with the community.

WHERE WE STAND

Feedback from the Nile-SEC team is positive. Wider feedback from country users is still pending: the regional training was postponed and is expected soon.

hydroserver.nilebasin.org, at a glance

8

Countries with data

32.9M

Observations stored

553

Datastreams

75

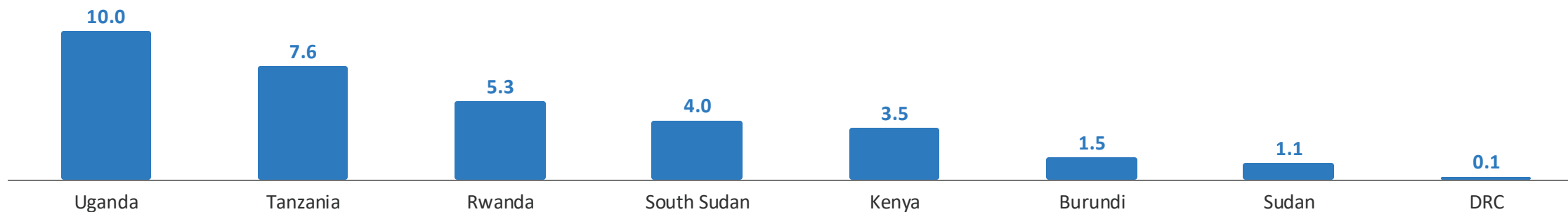
Monitoring sites

3.8 yrs

Of continuous data

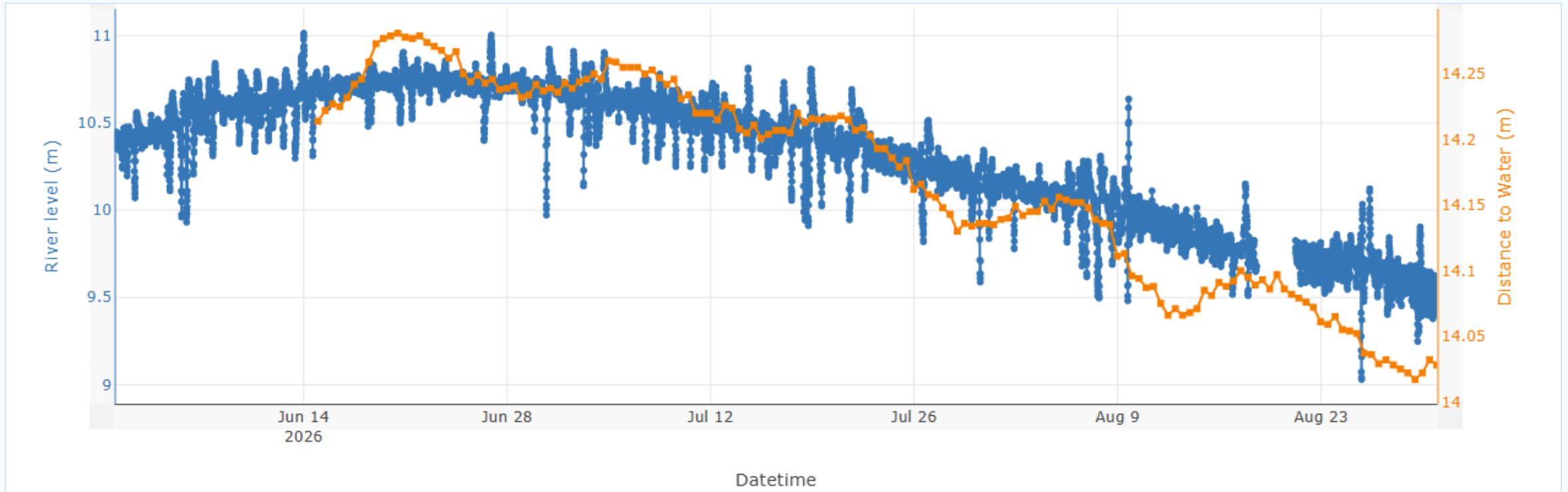
*Data spans 8 November 2022 through 15 September 2026, across Uganda, Tanzania, Rwanda, South Sudan, Kenya, Burundi, Sudan, and DRC
Egypt and Ethiopia have not yet onboarded.*

OBSERVATIONS BY COUNTRY WORKSPACE (MILLIONS)



Data visualization: reading datastreams together

HydroServer's visualization tool overlays datastreams on one time axis, so records from different stations and instruments can be read side by side.



Across locations

The same parameter at several stations, e.g. river level upstream and downstream on a shared river.

Across parameters

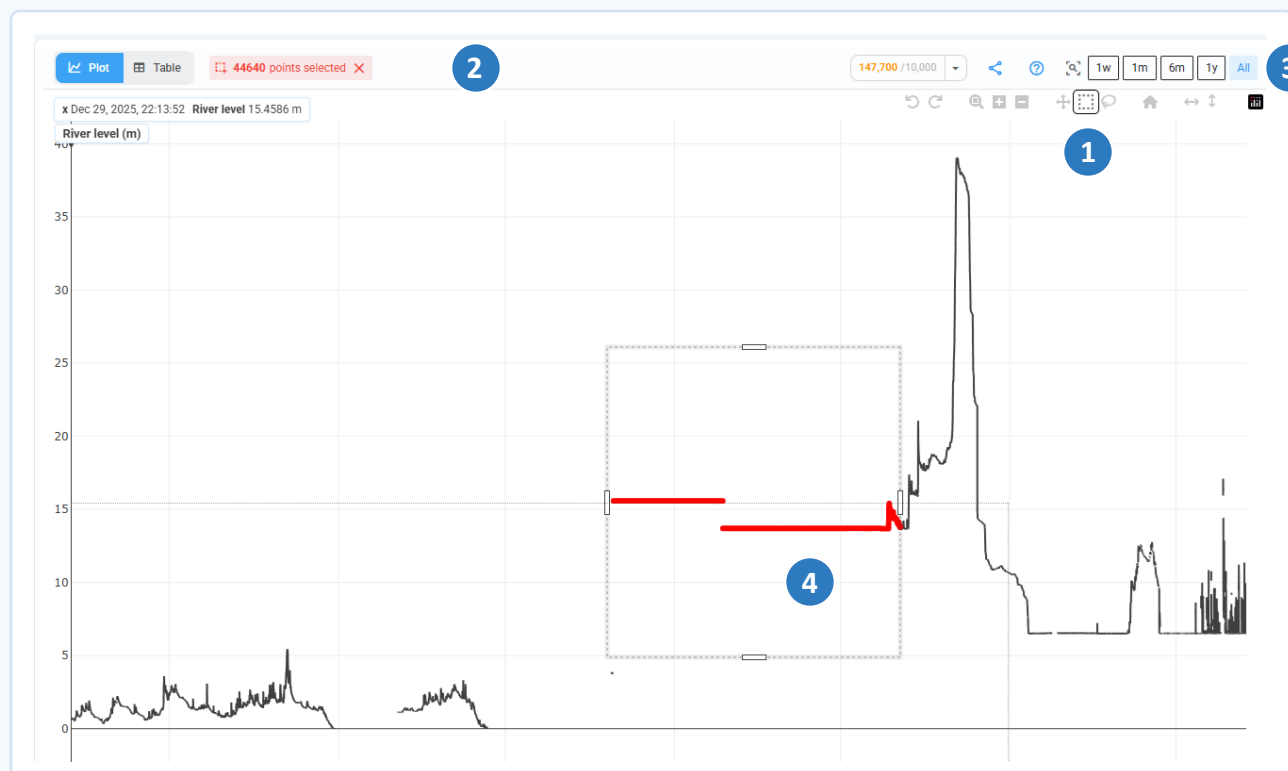
Different parameters at nearby sites, e.g. river level against groundwater depth in a nearby well (above).

WHY IT MATTERS FOR NBI

Rivers and aquifers cross borders. Seeing how they interact is evidence for transboundary water resources management.

Select it. Correct it. Keep the raw record.

Hydrologists clean their time series here every day: select exactly the points that are wrong, correct them, keep the raw record untouched.



SELECT EXACTLY THE DATA

- 1 Box and lasso selection**
Draw straight on the plot to pick the points you want.
- 2 Live count of the selection**
Always know how many points an edit will touch.
- 3 Zoom, pan and time windows**
1 week, 1 month, 6 months, 1 year or the full record; or a table view.
- 4 Selection highlighted**
The chosen segment turns red, ready to correct.

A real session: 147,700 points loaded, 44,640 selected, cursor at 15.4586 m on 29 Dec 2025.

We tested it on our own records: easy to use and very useful. It covers daily QC needs as well as commercial tools such as AQUARIUS, and open source, with no licence fee. We also tested the rating curve tool and generated discharge time series from river water level.

Downstream integration opportunities

HydroServer SensorThings API ^{1.1} OAS 3.1

/api/sensorthings/v1.1/openapi.json

This is the documentation for the HydroServer SensorThings API implementation.

Authorize 

Datastreams

GET	/api/sensorthings/v1.1/Datastreams	List Datastreams	 
POST	/api/sensorthings/v1.1/Datastreams	Create Datastream	
GET	/api/sensorthings/v1.1/Datastreams('{datastream_id}')	Get Datastream	 
PATCH	/api/sensorthings/v1.1/Datastreams('{datastream_id}')	Update Datastream	
DELETE	/api/sensorthings/v1.1/Datastreams('{datastream_id}')	Delete Datastream	

Features Of Interest

The open API, live

- hydroserver.nilebasin.org/api/sensorthings/v1.1/docs
- Interactive documentation — every endpoint browsable and testable

WHAT THIS OPENS UP

Any system that speaks REST/JSON — a national database, a forecasting model, a dashboard — can read regional data directly, with no custom adapter and no vendor contract.

The API also made it easy to **automate the migration** itself.

LOOKING AHEAD

The Nile Basin Digital Twin, including its AI Copilot, will build on existing components, drawing on the data this open API already serves.

Open API



AI Copilot

Nile Basin Digital Twin

What we would tell another users

Make the case on sustainability

DataSight did its job. The case was cost, control and extensibility, not failure.

Change only the regional node

Field stations, telemetry and national databases stayed as they were. That kept the risk low.

Model before volume

Get Sites, Sensors and Units right before uploading a single value.

Three verifiable tracks

Metadata, historical backfill, then cutover, station by station, only once verified.

Raw and QC apart from day one

Separate Datastreams for raw and corrected data cost far less than retrofitting later.

Open source is people, too

Work with the developer community, and give back: fixes, lessons, and ideas like branding in settings.

Community support: the key to a successful migration



HydroServer is open-source software, and its community shares more than code. The support of these people was instrumental to the success of NBI's deployment and migration.



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Utah State University



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Daniel Slaugh

Utah State University



Washington

World Meteorological Organization

THE OPEN-SOURCE SPIRIT

Shared code is only half of open source.
The other half is people willing to share their time and knowledge.

Open code

NBI built on the public HydroServer repository instead of starting from scratch.

Shared expertise

Guidance during the deployment process.

Giving back

NBI's lessons and improvements can flow back to the community.

Thank you

Questions and discussion